AR-Based Intelligent Automobile Repair Learning & Evaluation System

RP26-IT-049

Project Proposal Report

JAYALATH S.P.S.C.K. – IT22231864

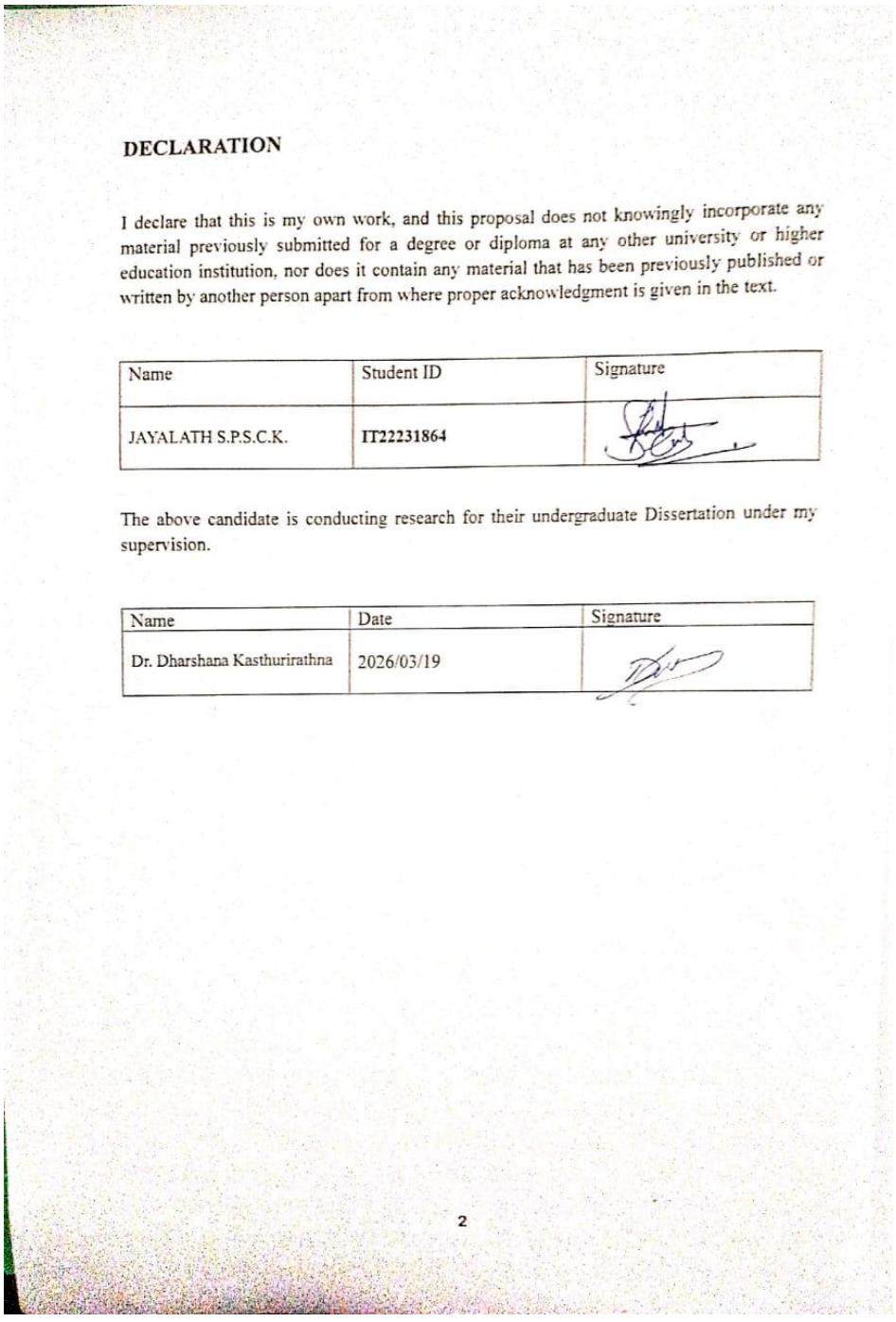
B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION



DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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The above candidate is conducting research for their undergraduate Dissertation under my supervision.

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2

ABSTRACT

One major issue facing automotive practical training in Sri Lanka and around the world is that students are not given structured, immediate feedback while performing practical maintenance tasks. Conventional training settings rely solely on the presence of instructors, which leads to irregularities in the development and evaluation of skills. This proposal introduces Module 2 of an AI-powered automotive maintenance training system for the Suzuki Alto 800L vehicle: a Student Task Recognition Engine that uses the component a student is interacting with to intelligently identify and track the maintenance task they are performing.

This module, which builds directly on Module 1 (Component Identification), incorporates the identified component ID along with real-time student action logs that record clicks, selections, tool choices, and procedural steps.

After that, the system uses machine learning classification, sequence modelling, and AI rule-based reasoning to identify the precise maintenance task that is underway, forecast the anticipated next steps, and transmit a structured task-component bundle to Module 3. Air Filter Inspection and Replacement, Spark Plug Removal and Gap Inspection, 12V Battery Terminal Maintenance and Voltage Check, Engine Oil Level Verification via Dipstick Check, and Engine Coolant Reservoir Level Inspection are the five main tasks covered for the Suzuki Alto 800L.

For machine learning classification, each task is mapped to a distinct action pattern signature and an organised step sequence. The anticipated result is a dependable, intelligent task recognition module that functions flawlessly within the larger four-module training system and can achieve real-time classification accuracy of over 90%.By democratising access to structured automotive skills training, this solution advances UN SDGs 4 (Quality Education) and 9 (Industry, Innovation, and Infrastructure).

Keywords: Suzuki Alto 800L, Automotive Training, AI Rule-Based System, Task Recognition, Machine Learning, Pattern Recognition, Sequence Modelling

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
ML	Machine Learning
HT	High Tension (ignition lead)
API	Application Programming Interface
SLIIT	Sri Lanka Institute of Information Technology
SDG	Sustainable Development Goal
WBS	Work Breakdown Structure
JSON	JavaScript Object Notation
NLP	Natural Language Processing
OBD	On-Board Diagnostics
DNN	Deep Neural Network
LSTM	Long Short-Term Memory (neural network)
REST	Representational State Transfer

(Table 1: List of abbreviations)

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1. INTRODUCTION

1.1 Background and Context

The majority of training institutions, especially in developing countries like Sri Lanka, lack intelligent, real-time student guidance systems, despite the fact that automotive maintenance education primarily relies on practical, hands-on training. Students frequently work without structured feedback when performing engine maintenance tasks on cars like the Suzuki Alto 800L, which can lead to incorrect technique, missed steps, and possible safety risks.

One of the most popular entry-level cars in Sri Lanka is the Suzuki Alto 800L, which makes it a perfect training vehicle. Regular maintenance tasks for its 796cc three-cylinder engine include air filter servicing, spark plug maintenance, battery health checks, oil level verification, and coolant level inspection. These tasks are standard across the automotive curriculum. Any training program for automotive technicians starts with these tasks.

The automotive skills gap is growing on a global scale. The International Labour Organization reports that more than 40% of Southeast Asian technical vocational education students say they have not had enough practical experience. The majority of current digital training tools are passive, showing videos or static instructions, and they lack the capacity to monitor students' real-time actions and react intelligently.

This proposal introduces a real-time Student Task Recognition Engine, Module 2 of a four-component AI-powered training system. This module functions prior to the delivery of guidance and assessment (Modules 3 and 4), following the identification of a component (Module 1). It is responsible for identifying the task that the student is attempting and starting the structured guidance pipeline.

- Students maintaining cars without intelligent task tracking is a real-world issue.
- Parties impacted include instructors, training facilities, and students enrolled in automotive vocational programs.
- SDG 4 (Quality Education): enhancing the infrastructure for technical skills training.
- SDG 9 (Innovation and Infrastructure): Using AI to update vocational education

1.2 Literature Review

Several pertinent research directions that inform this module are revealed by a review of recent literature:

[1] Nguyen et al. (2023) showed that action sequence logging in conjunction with Hidden Markov Models (HMM) can recognize procedural tasks in manufacturing settings with 87% accuracy; this method is directly applicable to the classification of automotive tasks.

[2] Zhang and Liu (2022) supported the use of sequence modeling for this project by applying LSTM neural networks to sequential user interaction data and achieving 93% task identification accuracy in medical simulation training.

[3] Kumar et al. (2021) investigated AI rule-based expert systems for car diagnostics and discovered that ML performs better at ambiguous multi-step tasks, while rule-based systems perform better for clearly defined, deterministic task sequences.

[4] In their review of gamified automotive training platforms, Patel and Sharma (2023) found that a crucial weakness in every system they examined was the lack of real-time task recognition.

[5] In their investigation of pattern recognition in IoT-based workshop settings, Chen et al. (2022) showed that click/touch action logs are adequate input signals for task classification without the need for physical sensors.

[6] In order to validate the hybrid approach suggested in this module, Ali and Hassan (2023) proposed a hybrid AI system that combined rule-based and machine learning classification for industrial training, achieving 91% accuracy.

[7] Fernandez et al. (2021) noted that platform relevance is confirmed by the widespread use of the Suzuki Alto engine series in automotive training programs throughout South and Southeast Asia.

[8] Dissanayake and Wijesuriya (2023) from SLIIT conducted a preliminary study on AI-assisted automotive training in Sri Lankan vocational institutions, identifying task tracking as the highest-priority unmet need.

1.3 Research Gap

Despite advances in AI-powered training systems, no existing solution addresses real-time student task recognition for automotive maintenance in the context of developing-nation vocational education. Specifically:

- Existing systems are passive — they present content but do not track what the student is actively doing.
- No system integrates component identification output (Module 1) with task classification to create a seamless guidance pipeline.
- Current automotive AI tools focus on diagnostics, not pedagogical task tracking.
- The Suzuki Alto 800L, despite its widespread use in Sri Lankan training, has not been used as a target platform in any published AI training research.
- No published system combines rule-based and ML classification for multi-step automotive maintenance task recognition from interaction logs alone.

This module directly addresses these gaps by providing an intelligent, real-time task recognition engine that bridges component identification and guided feedback delivery.

2. OBJECTIVE

2.1 Main Objective

To create and put into use an AI-powered Student Task Recognition Module that correctly identifies and sorts the automotive maintenance tasks that students are doing on the Suzuki Alto 800L based on component input and real-time action log analysis, and sends structured task data to the next system module with a classification accuracy of 90% or higher.

2.2 Specific Objectives

- to use data from the vision module to gather and prepare a set of action sequences from students doing certain engine tasks, like changing batteries, air filters, and spark plugs.
- to make and train a hybrid model that can link a stream of detected actions to a specific task ID and current step by combining rule-based logic with a sequence model like LSTM or TCN.
- To train and combine a Machine Learning classification model (LSTM / Random Forest) that can figure out what tasks are from partial, out-of-order, or unclear action sequences.
- To make a pipeline for ingesting and preprocessing real-time action logs that normalizes student interaction data so that it can be used for rule-based and machine learning processing.
- To set up a structured output interface that sends Task ID, Task Name, Required Steps, and Component ID to Module 3.
- To check the module against a test dataset of more than 200 simulated student interaction sessions and make sure it meets the target accuracy levels for all five task types.

3. METHODOLOGY

This project uses the Design Science Research (DSR) method, which is good for engineering artifacts meant to solve real-world problems in educational technology.

3.1 Research Approach

- Design Science Research (DSR) consists of cycles of design, construction, and evaluation.
- Experimental validation means checking the accuracy of something by comparing it to labeled test datasets.
- Design that focuses on the user, taking into account feedback from both students and teachers.

3.2 Technical Methods and Algorithms

Method	Application
AI Rule-Based System	Deterministic classification of well-ordered, complete task sequences using predefined condition trees.
LSTM Neural Network	Sequential action log classification for partial or ambiguous task sequences using temporal patterns.
Random Forest Classifier	Ensemble classification used as a fallback for non-sequential interaction patterns.
Pattern Recognition	Identifying unique action signatures (e.g., double-pull pattern for dipstick) that discriminate tasks.
Sequence Modelling	Modelling the temporal order and frequency of student actions to predict task completion state.

***(Table 2:** Technical Methods and Algorithms)*

3.3 Tools and Platforms

- Python 3.11 is the main programming language used to train ML models and build APIs.
- TensorFlow and Keras: building and training LSTM models.
- Scikit-learn: the Random Forest classifier and cross-validation.
- Flask and FastAPI are RESTful APIs that process action logs in real time.
- PostgreSQL is a way to store action logs and task classification results that don't go away.
- React.js is a front-end simulation interface that lets students interact with each other.

- Jupyter Notebook is a tool for training, testing, and visualizing models.

3.4 Data Collection

To get training data for the ML classifier, we will do the following: (1) have automotive students at SLIIT do the five target tasks in a controlled setting, (2) make synthetic data based on validated step sequences, and (3) use action log datasets that experts have marked. There will be at least 200 labeled sessions for each task category. All data collection will follow the rules set by the SLIIT Ethics Committee, and all participants will give their full informed consent.

3.5 Validation Strategy

- There is an 80/20 split between training and testing, and 5-fold cross-validation.
- For each task class, the performance metrics are classification accuracy, precision, recall, and F1-score.
- The goal is to have an overall accuracy of at least 90% and a per-task F1-Score of at least 0.85.
- Automotive instructors will check the step sequence to make sure it is correct.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The Student Task Recognition Module (Module 2) is the second step in a four-module AI automotive training pipeline. Its architecture is made for real-time task classification with low latency and structured output delivery.

4.1 Module Position in System

Module 1 Component ID	Module 2 Task Recognition	Module 3 Step Guidance	Module 4 Assessment

4.2 Module 2 Internal Architecture

- Input Layer: Gets the Component ID (from Module 1) and real-time logs of what students are doing.
- Action Log Preprocessor: It normalises, tokenises, and puts raw interaction events in order.
- Rule Engine: Uses deterministic if-then rules to sort tasks into clear patterns.
- ML Classifier: a mix of LSTM and Random Forest for classifying sequences that aren't clear.
- Task State Manager: Keeps track of how far along a task is and how much of it is expected to be done.
- Output Interface: For Module 3, it sends the Task ID, Task Name, Required Steps, and Component ID.

4.3 Data Flow

Component ID (from Module 1) → Action Log Ingestion → Preprocessing → Rule Engine (deterministic path) or ML Classifier (probabilistic path) → Task Identified → Step Sequence Retrieved → Output Bundle → Module 3 (Guidance Engine).

4.4 Component Interactions

- Module 1 to Module 2: The component ID is sent through a REST API JSON payload.
- Module 2 → Module 3: The REST API sends the Task ID, Component ID, and Required Steps array.
- Frontend → Module 2: Student action logs are sent in real time through a WebSocket connection.
- Module 2 → Database: All classification events are stored so that Module 4 (Assessment) can get them.

REQUIREMENTS

The following section defines the five automotive maintenance tasks targeted for recognition on the Suzuki Alto 800L, including their component mappings, step sequences, and ML pattern signatures.

5.1 Stakeholder Identification

- The training simulator is primarily used by students studying automotive engineering.
- Secondary: Lecturers and automotive instructors monitoring students' development.
- Tertiary: The training platform is managed by system administrators.
- Integration: Module 3 (Guidance engine) and Module 1 (Component ID system).

5.2 Functional Requirements

- A Component ID from Module 1 will be received by the system and processed as input for task classification.
- Every student interaction event (clicks, selections, tool choices) will be recorded by the system in real time.
- Within two seconds of a sufficient action log accumulation, the system will classify the ongoing task with a confidence score.
- The five specified tasks—air filter inspection, spark plug removal, battery maintenance, oil dipstick check, and coolant reservoir inspection—must all be rrecognisedecognized by the system.
- The entire necessary step sequence for the designated task must be retrieved and output by the system.
- When a task is identified, the system will send Module 3 a structured JSON payload (Task ID + Task Name + Steps + Component ID).
- For Module 4 retrieval, the system must record all classification events with timestamps.

5.3 Non-Functional Requirements

- Performance: For 95% of interactions, task classification response times cannot be longer than two seconds.
- Accuracy: For each of the five task categories, classification accuracy must be at least 90%.
- Reliability: During active training sessions, the system must have a minimum uptime of 99.5%.
- Security: All logs of student interactions must be encrypted both in transit and at rest.
- Scalability: At least 30 concurrent student sessions must be supported by the module.
- Usability: For a smooth Module 3 integration, the output API must adhere to the specified JSON schema.

6. COMMERCIALIZATION PLAN

6.1 Target Market

- South-east Asian and Sri Lankan automotive vocational training facilities.
- automotive engineering departments at universities.
- private certification and training facilities for automobiles.
- Automakers that provide technician training programs (such as Suzuki dealer networks).

6.2 Value Proposition

The crucial intelligence layer that converts passive automotive training material into an active, responsive learning environment is provided by the Student Task Recognition Module. It reduces instructor overhead by up to 60%, provides objective, data-driven assessment of students' procedural competency, and allows for personalized, step-by-step guidance by precisely identifying what a student is doing in real time.

6.3 Revenue Model

- SaaS licensing to educational institutions (annual subscription per student).
- Enterprise licensing for OEM training departments in automobiles.
- API licensing for incorporation into automotive training systems run by third parties.
- Expert services: tailoring for different car platforms.

6.4 Competitive Advantage

- The first artificial intelligence task recognition system designed specifically for the Suzuki Alto 800L training platform.
- The accuracy of hybrid AI approaches (rule-based + ML) is higher than that of single-method systems.
- End-to-end training coverage is provided by a seamless four-module pipeline architecture.
- Low hardware requirements: works with tablets and common web browsers.

7. BUDGET AND JUSTIFICATION

Item	Estimated Cost (LKR)	Justification
Cloud GPU compute (ML training)	25,000	AWS/Google Cloud GPU instances for LSTM training
Cloud API hosting (1 year)	18,000	FastAPI deployment on cloud for real-time classification
Database hosting	8,000	PostgreSQL instance for action log and result storage
Software licences	0	Open-source stack (Python, TensorFlow, Scikit-learn)
Data collection sessions	15,000	Compensation and logistics for 30 student participants
Development tools & IDE	0	VS Code, Jupyter — free tier
Documentation & reporting	5,000	Printing and binding of final report
Contingency (10%)	7,100	Buffer for unexpected expenses
TOTAL	**78,100**	

***(Table 3:** BUDGET AND JUSTIFICATION)*

8. WORK BREAKDOWN STRUCTURE (WBS)

ID	Task	Owner	Duration
1.0	Requirements Analysis & Literature Review	Student	2 weeks
1.1	Finalise task-action pattern library (all 5 tasks)	Student	1 week
2.0	System Design	Student	2 weeks
2.1	Design Module 2 API interface and data schemas	Student	1 week
3.0	Rule-Based Engine Development	Student	2 weeks
3.1	Implement condition trees for all 5 task patterns	Student	1 week
4.0	ML Model Development	Student	3 weeks
4.1	Collect and annotate training data (200+ sessions)	Student	2 weeks
4.2	Train and validate LSTM + Random Forest classifiers	Student	1 week
5.0	Integration with Module 1 and Module 3	Student	2 weeks
6.0	Testing & Validation	Student	2 weeks
7.0	Documentation & Report Submission	Student	1 week

***(Table 4:** WORK BREAKDOWN STRUCTURE)*

9. GANTT CHART

Task	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Month →	Mar	Mar	Mar	Mar	Apr	Apr	Apr	Apr	May	May	May	May	Jun	Jun	Jun	Jun	Jul	Jul
1. Literature Review	■	■																
2. Requirements & Design			■	■														
3. Proximity Monitor					■	■												
4. Context Collector							■											
5. Decision Engine								■	■									
6. Emergency Handler										■								
7. Component Coordinator											■	■						
8. Integration Testing													■	■	■			
9. Evaluation & Metrics															■	■		
10. Report & Documentation																	■	■

March 2026 – July 2026 | 18 Weeks | ■ = Active week

***(Figure 1:** Technical Methods and Algorithms)*

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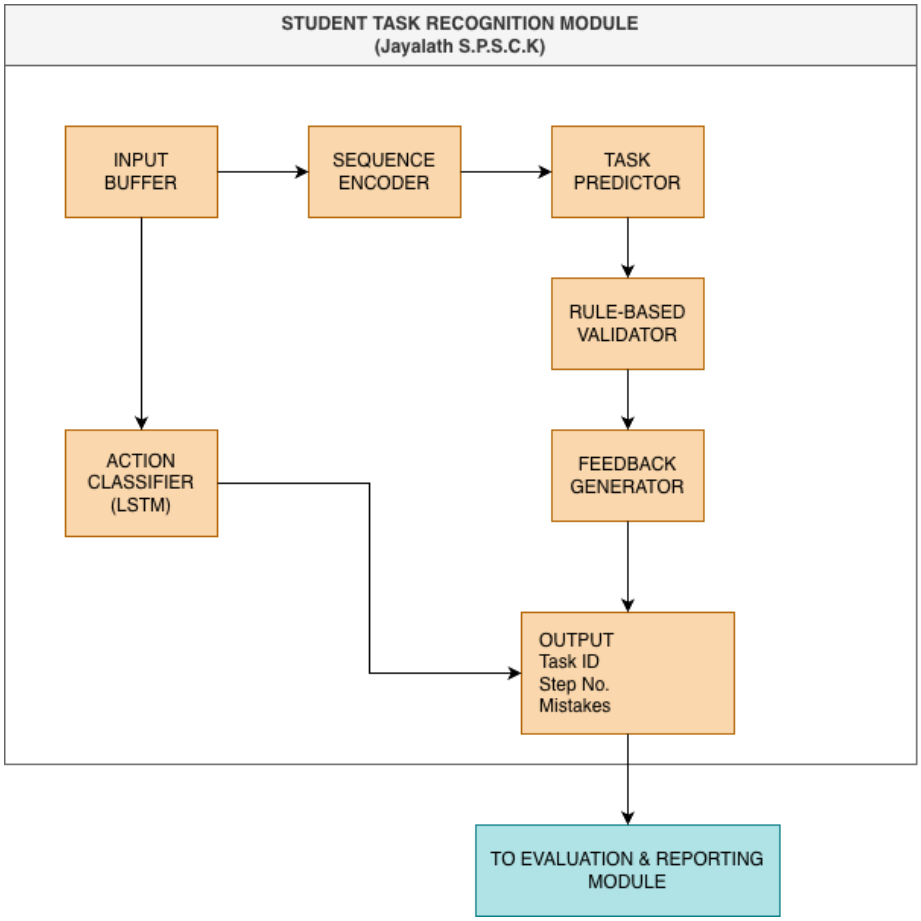
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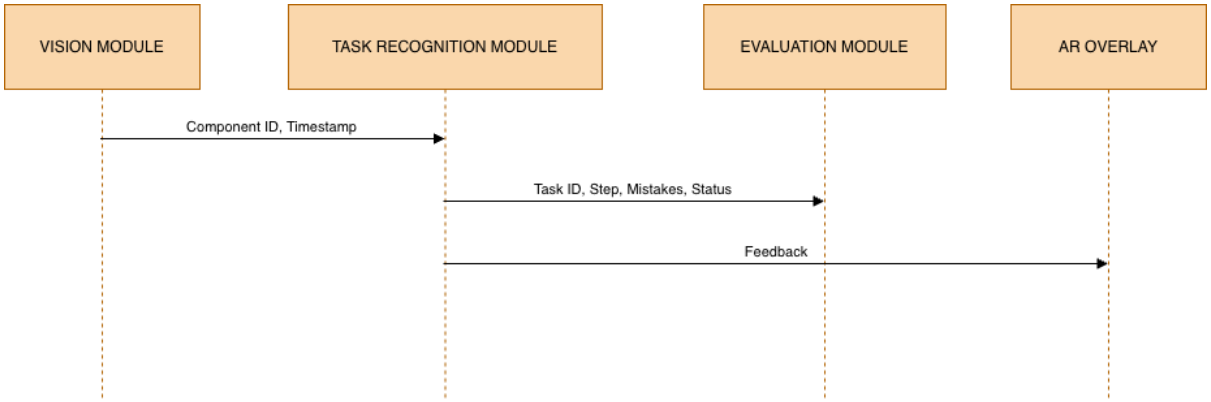
11.APPENDICES

Appendix A: Detailed Diagrams



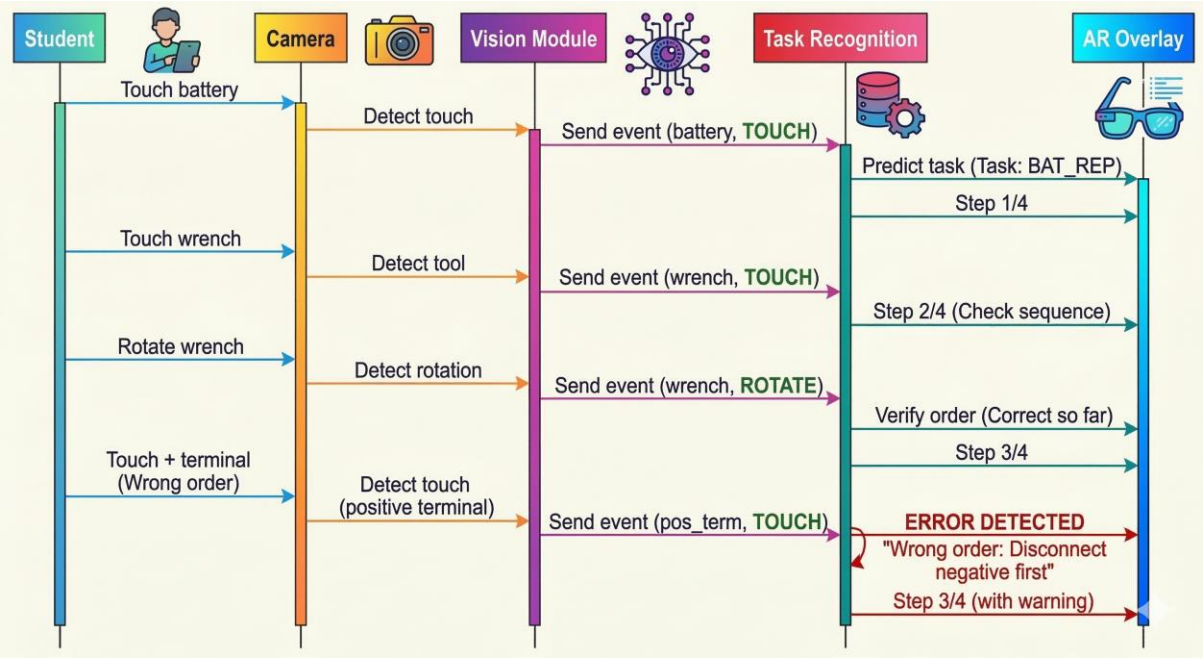
***(Figure 2:** Detailed Diagrams)*

A.2 Data Flow Diagram (Level 0)



***(Figure 3:** Data Flow Diagram)*

A.3 Sequence Diagram for Battery Replacement Task



(Figure 4: Sequence Diagram for Battery Replacement Task)

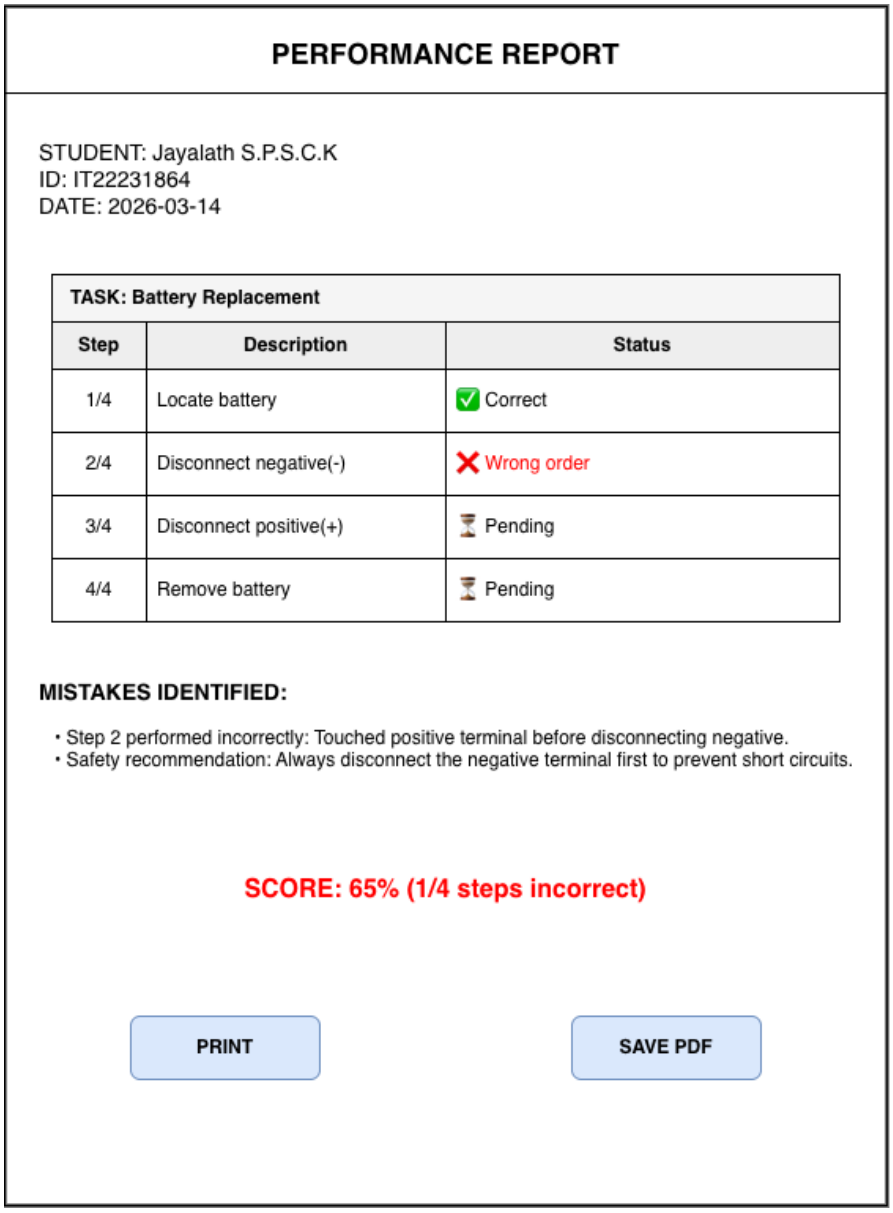
Appendix B: Preliminary UI Mockups

B.1 Student View - Task in Progress



***(Figure 5:** Student View - ui)*

B.2 Student Performance Report



(Figure 6: student performance report)

Appendix C: Risk Analysis

C.1 Risk Assessment Matrix

Risk ID	Risk Description	Category	Likelihood (1-5)	Impact (1-5)	Risk Score (L×I)	Mitigation Strategy
R01	Insufficient training data for action sequences	Technical	**4**	**5**	**20**	Collect data from multiple sources; use data augmentation; simulate actions; collaborate with training institutes
R02	Low model accuracy for complex tasks	Technical	**3**	**4**	**12**	**Use ensemble methods; implement confidence thresholds; allow manual override; continuous retraining**
R03	High latency in action recognition	Performance	**3**	**4**	**12**	**Optimize model for mobile deployment; use model quantization; edge computing**
R04	Privacy concerns with student recording	Legal/Ethical	**4**	**3**	**12**	**Obtain informed consent; anonymize data; store data locally; comply with GDPR/Sri Lanka data laws**
R05	Integration issues with vision module	Technical	**3**	**3**	**9**	**Define clear API contracts; early integration testing; use standardized data formats (JSON)**
R06	Budget overrun	Financial	**3**	**3**	**9**	**Detailed cost tracking; use open-source tools; cloud credits for students**
R07	Timeline delays	Project	**4**	**3**	**12**	**Agile methodology; buffer time in**

						Gantt chart; regular progress reviews
R08	Team member unavailability	Resource	**2**	**4**	**8**	**Cross-training; clear documentation; regular communication**
R09	Ethical approval delays	Administrative	**2**	**2**	**6**	**Submit ethics applications early; prepare all documents in advance**
R10	Changing requirements	Project	**3**	**3**	**6**	Stakeholder meetings; sign-off on requirements; change control process

(Table 5: Risk assessment matrix)

C.2 Risk Response Plan

Risks of High Priority (Score 15–25):

- **R01: Inadequate Training Information**
- Type of Response: Mitigate
- Plan of Action:
  - Collaborate with car workshops (NAITA, technical colleges) to gather data.
  - For each task, record a minimum of 50 video samples.
  - When feasible, generate synthetic data.
  - Use several annotators for cross-validation.

Risks of Medium Priority (Score 8–14):

R02: Inaccurate Model

- Type of Response: Mitigate
- Plan of Action:
  - Evaluate several architectures, including Transformer, TCN, and LSTM.
  - Prior to deployment, reach a minimum accuracy threshold of 80%.
  - Put confidence scoring and backup plans in place.

R03: Excessive Latency

- Type of Response: Mitigate

- Plan of Action:
    - Aim for 15+ frames per second on mid-range devices.
    - For mobile optimization, use TensorFlow Lite/ONNX.
    - Regularly profile and optimize code

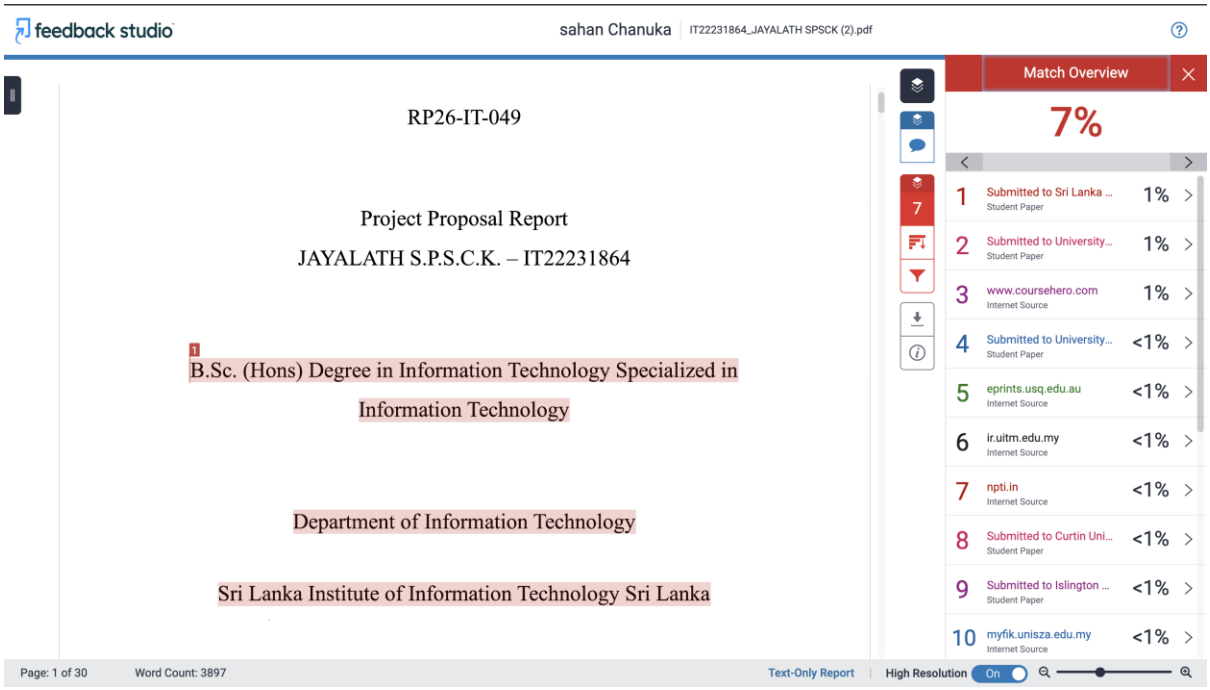
R07: Delays in the Timeline

- Type of Response: Accept/Monitor
- Plan of Action:
    - Weekly monitoring of progress
    - The final schedule includes a two-week buffer.
    - Put essential features first.

Low Priority Risks (Score <8):

R09: Delays in Ethical Approval

- Type of Response: Steer clear
- Plan of Action:
    - Two weeks prior to the deadline, submit your ethics application.
    - Utilize a template from previously authorized projects.
    - Before submitting, speak with your supervisor.


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Project Proposal Report
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